

Q2:

A- The dimensionality of the covariance matrix: $S \in \mathbb{R}^{2 \times 2}$

B-

$$\begin{bmatrix} -2 & -2 \\ 0 & 0 \\ 2 & 2 \end{bmatrix} \text{ Xcentered}$$

$$\begin{bmatrix} 8 & 8 \\ 8 & 8 \end{bmatrix} \quad X_{centered}^T @ \text{ Xcentered}$$

$$\begin{bmatrix} 4 & 4 \\ 4 & 4 \end{bmatrix} \quad \text{Final covariance matrix after multiplying with } 1/(N-1)=1/2$$

Q3:

A – s5: X1, X2 spread similarly around their zero-means

B – s2: X2 spreads more largely than X1

C – s1: X1 spreads more largely than X2

D – s3: $X_1 \text{ increases, } X_2 \text{ decreases} \rightarrow \text{Cov}(X_1, X_2) < 0$

E – s4: $X_1 \text{ increases, } X_2 \text{ increases} \rightarrow \text{Cov}(X_1, X_2) > 0$

Q4: See notebook

Q5:

A -

$$J(w) = \frac{w^T S_B w}{w^T S_W w} \quad (\text{original space}) = \frac{(m_2 - m_1)^2}{s_1^2 + s_2^2} \quad (\text{projected space}) \quad \leftarrow \text{maximize}$$

B - The numerator is the average distance (variance) between classes. The denominator is the average variance within classes. Maximizing this means preserving maximal relative distance between classes.

$J(w) \in \mathbb{R}$, scalar score measuring how well w separates the classes relative to their within class spread

$w \in \mathbb{R}^{D \times 1}$, vector that defines the direction onto which the data are projected.

$S_B \in R^{D \times D}$, total between-class covariance matrix

$S_W \in R^{D \times D}$, total within-class covariance matrix

$m_k \in R$, class mean vectors, projected quantities $\rightarrow$ scalars

$s_1, s_2 \in R$, variances of the two classes in the projected space $\rightarrow$ scalars

C – Yes, because the objective function $J(w)$ is differentiable, meaning a maximum scoring set of params can be calculated without iteration.

D – The data have to be distributed according to a gaussian distribution **and** every category has the same covariance matrix.

E – Supervised, as the data is labelled

F – For classification

Q6:

A –

$$w = S_W^{-1}(m_2 - m_1)$$

$$b = -0.5 w^T (m_1 + m_2)$$

B –

$w \in R^1$ (scalar), because the data only has one dimension.

C –

$$w = S_W^{-1}(m_2 - m_1) = \frac{3.5 - 1.5}{\frac{1}{2}[(1 - 1.5)^2 + (2 - 1.5)^2 + (3 - 3.5)^2 + (4 - 3.5)^2]} = \frac{2}{0.5} = 4$$

$$b = -0.5 w^T (m_1 + m_2) = -2(1.5 + 3.5) = -10$$

Q7:

A – $g(X) = X w + b$, $\hat{y} = [1 \text{ if } g(x) > 0, \text{ else } 0]$

B – $\hat{y} \in N^{N \times 1}$

C – $g(X) = X w + b = [5 \ 11 \ 17] + 3 = [8 \ 14 \ 20]$ So the labels are: $\hat{y} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$

Q8:

A – You must reshape the data matrix X into a flattened vector. That is the only way the LDA can read it. Now every channel/time interval combination is a feature.

$$B - g(x) = w^T x + b = [1 \ -1 + 3 \ 1 + 1 \ -1 + 2 \ -1 + 3 \ 2 + 2 \ -1] + 2 = 5 = \text{positive so the label is } \hat{y} = 1$$

Q9: See notebook

Q10:

Contribution

Everyone made the exercises and notebooks individually/fully, I (Robert) merged the answers together, which we all compared during the workgroup. Thanasis and I (Robert) did the exam questions.

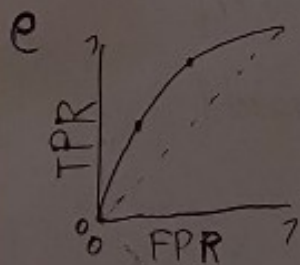
Q11:

$$c) g(x_0) = \vec{w}^T \vec{x} \quad \text{where } \vec{w} = \begin{bmatrix} -3 \\ 4 \\ -2 \end{bmatrix}, \quad \vec{x} = \begin{bmatrix} 1 \\ -7 \\ -3 \end{bmatrix}$$

$$g(x_0) = [-3 \ 4 \ -2] \begin{bmatrix} 1 \\ -7 \\ -3 \end{bmatrix} = -3(1) + 4(-7) - 2(-3) = -3 - 28 + 6 = -25$$

$$g(x_0) = -25 \leq 0 \Rightarrow \text{class } 0$$

d	y	g(x)	g(x)-b ₁	g(x)-b ₂	
	1	0.95	0.25 TP	0.45 TP	TPR _{b1} = $\frac{2}{2+2} = \frac{1}{2}$
	1	0.85	0.15 TP	0.35 TP	FPR _{b1} = $\frac{1}{1+3} = \frac{1}{4}$
	1	0.65	-0.05 FN	0.15 TP	TPR _{b2} = $\frac{3}{3+1} = \frac{3}{4}$
	1	0.45	-0.25 FN	-0.05 FN	FPR _{b2} = $\frac{2}{2+2} = \frac{1}{2}$
	0	0.20	-0.10 FP	0.30 FP	
	0	0.60	-0.10 FNTN	0.10 FP	
	0	0.75	-0.55 FNTN	-0.35 TN	
	0	-0.75	-0.85 TTN	-0.65 TN	



Same true and false positive rate = random policy score?

g No, a linear transformation only moves $g(x)$ Since $a > 0$, if $f(x) > 0 \rightarrow af(x) > 0$

a will not change the ROC curve

~~However, changing b can (as seen in d)~~

Changing b will only change where you land on the curve

Cov

20-21 resit 1d

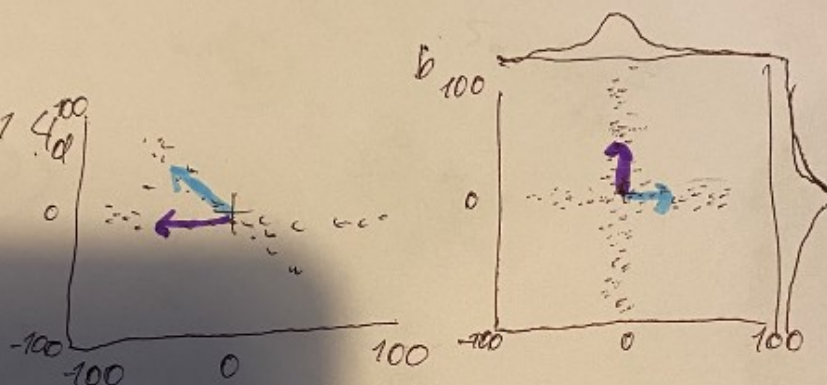
$a: \Sigma_4$ $b: \Sigma_3$ $c: \Sigma_1$ $d: \Sigma_2$

21-22 resit 1d

$S_1 \neq S_2$, and C_2 isn't Gaussian

LDA

20-21



$W \in \mathbb{R}^{D \times D}$



25-263

$$a) W = \sum (N_{01} - N_{00}), b = -\frac{W^T(N_{00} + N_{01})}{2}$$

$$b = \frac{-1}{2} [3(8) - 2(7)] = -5$$

$$b) W = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 8-2 \\ 4-3 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 4 \\ 1 \end{bmatrix} = \begin{bmatrix} 1(4) - 1(1) \\ -1(4) + 2(1) \end{bmatrix} = \begin{bmatrix} 3 \\ -2 \end{bmatrix}$$

h 1: The sample used for the calculation could be not representative of the population.

2: Depending on N , Covariance isn't normalized

i, ideq: Introducing bias often lowers variance

